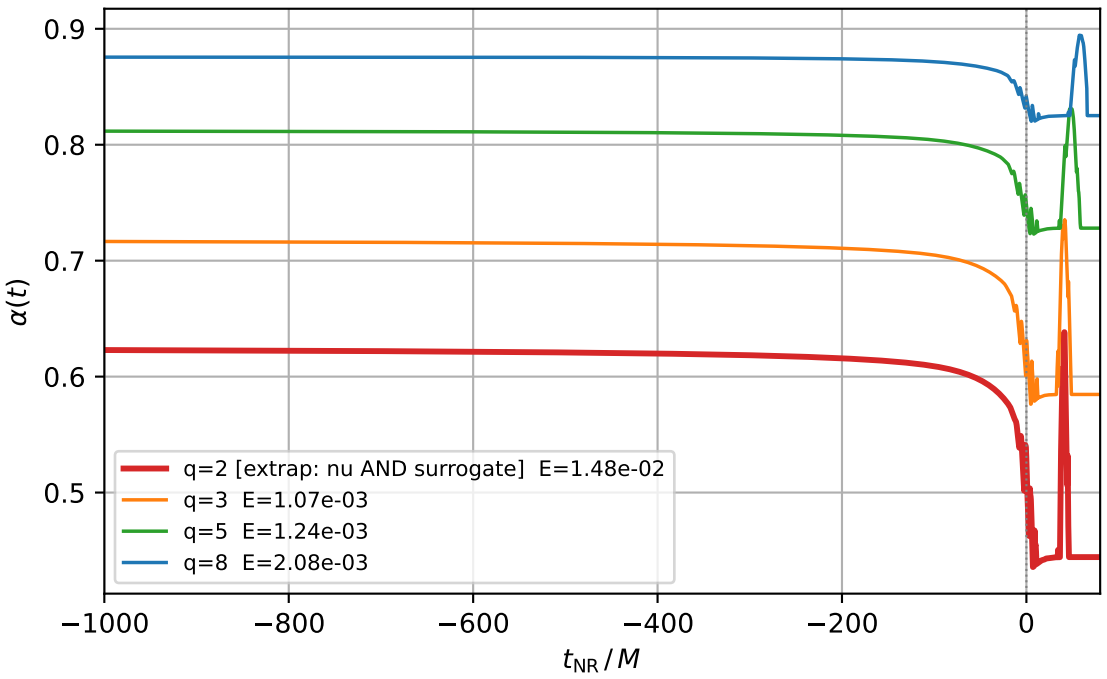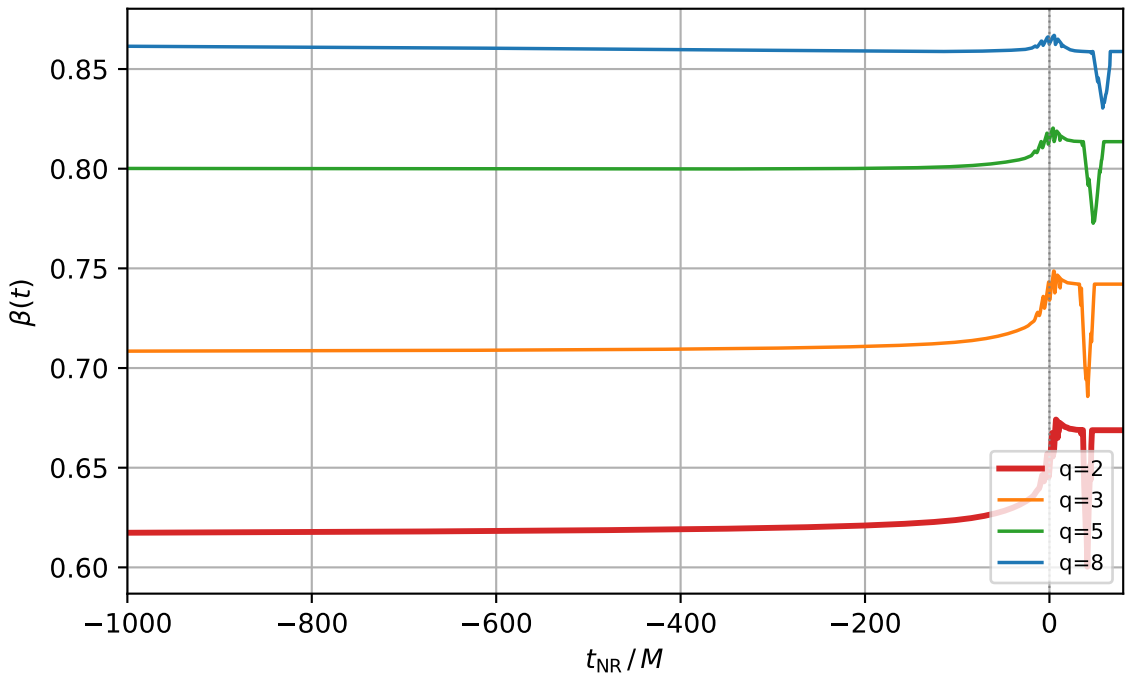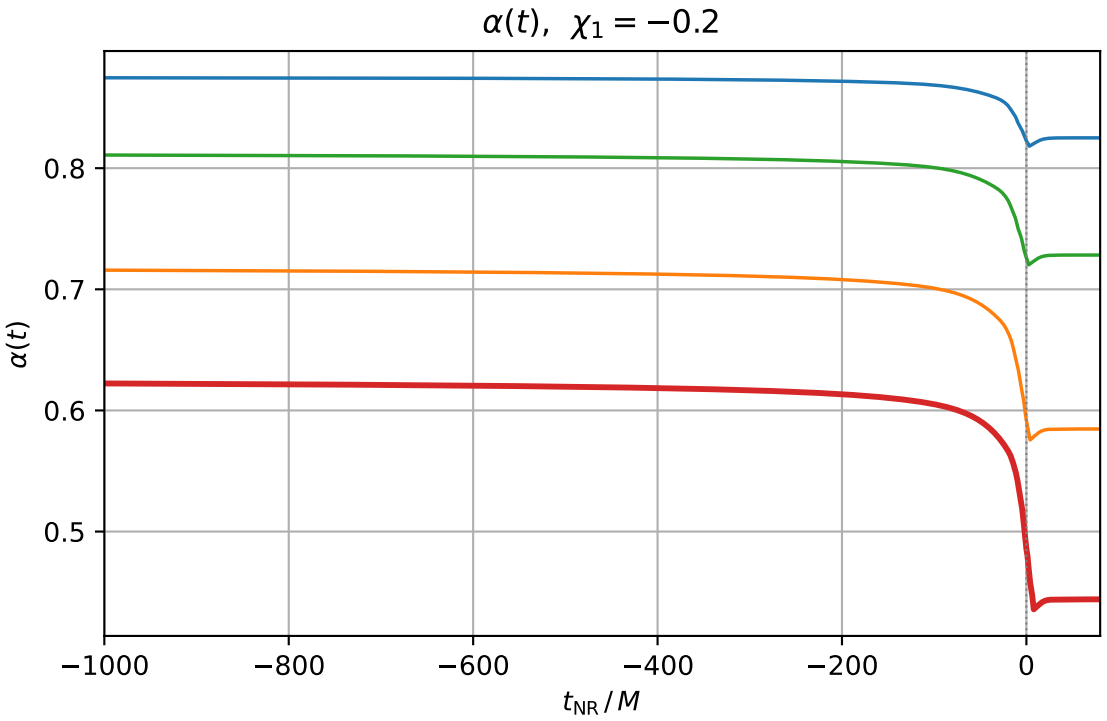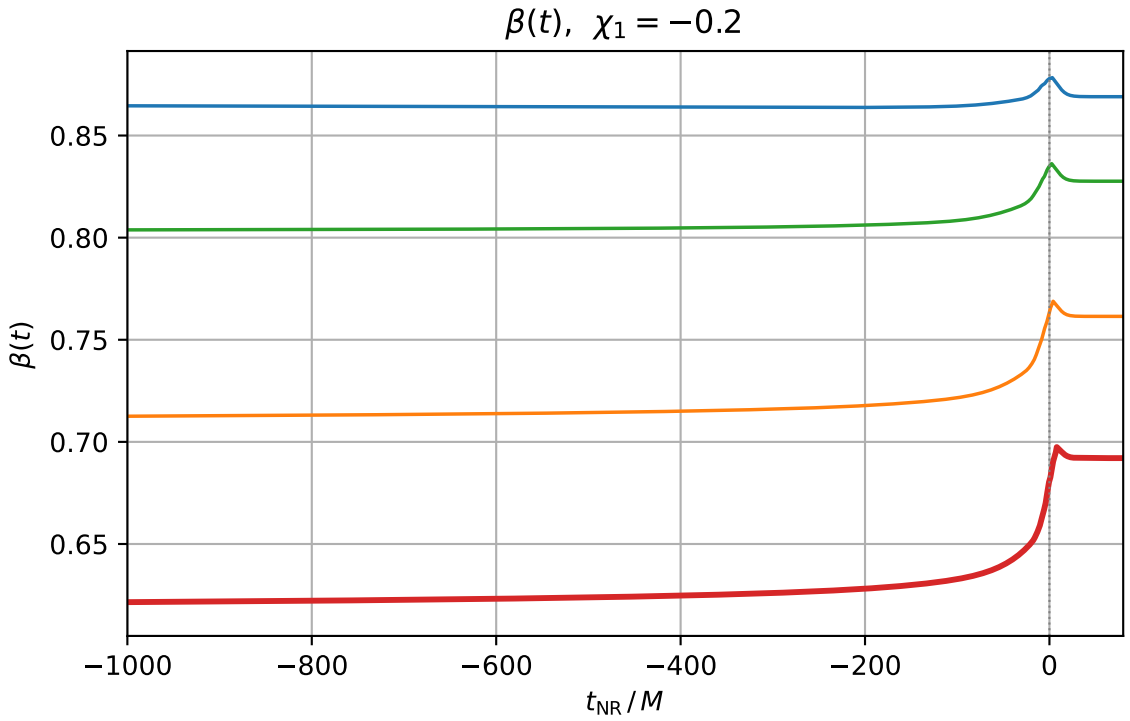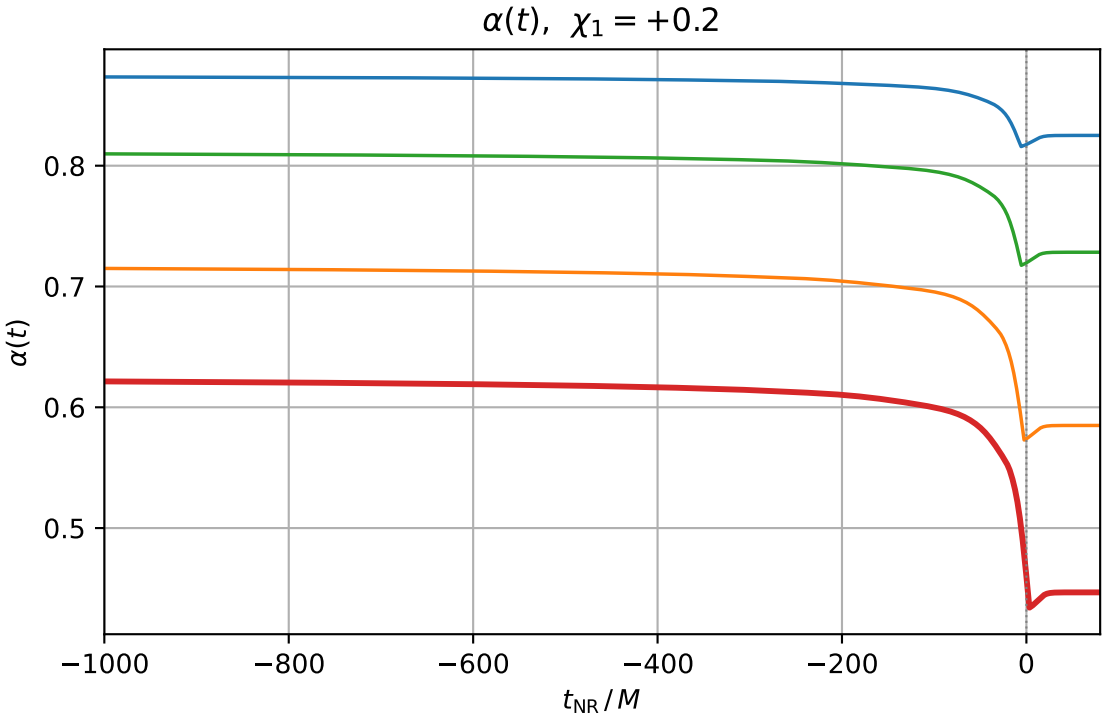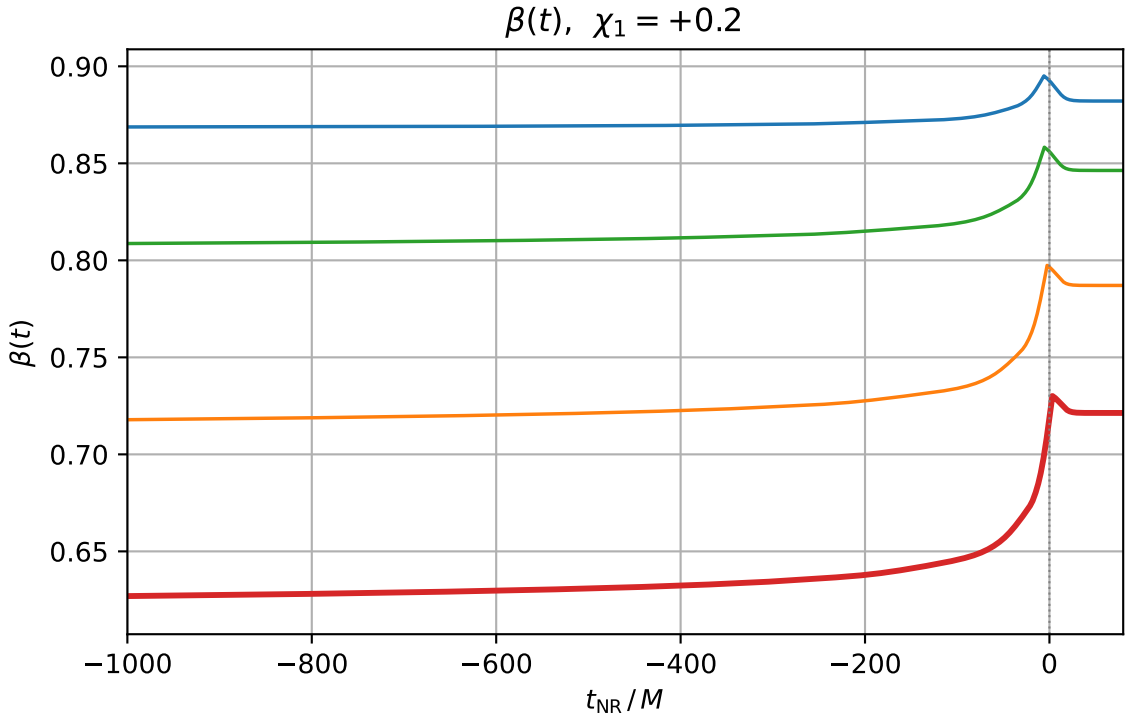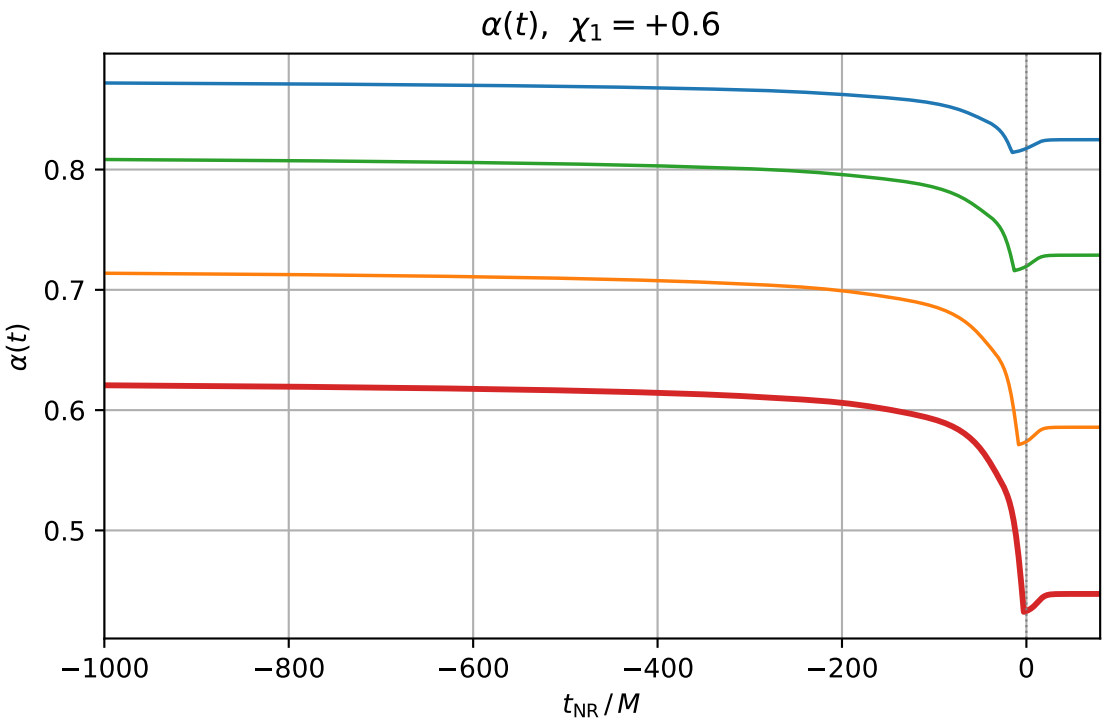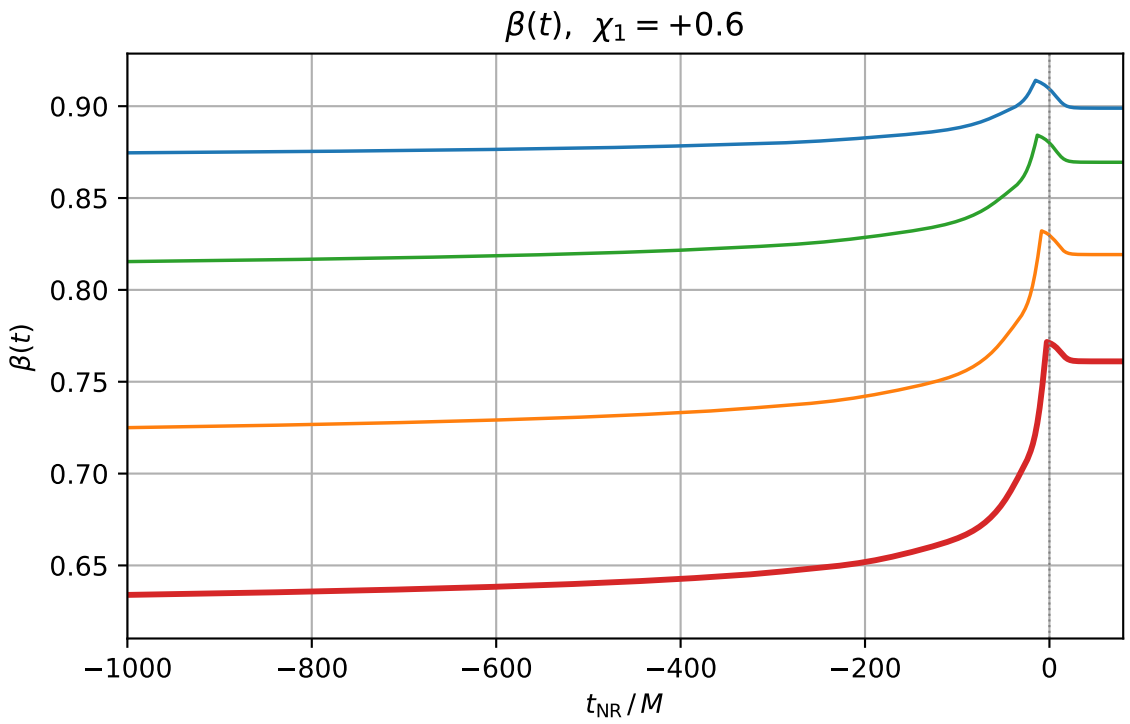pn_anc_spin_physical: master α/β, $\chi_1 \in \{-0.6, -0.2, +0.2, +0.6\}$